Lesson 4: Sample Integration & Comparative Analysis

Spatial Transcriptomics Course

Learning Objectives

- Integrate multiple spatial transcriptomics samples
- Remove technical batch effects while preserving biology
- Identify differentially expressed genes between conditions
- Perform gene set enrichment analysis (GSEA) with KEGG
- Compare cell type proportions across conditions
- Visualize treatment effects in spatial context
- Align tissue slices to a shared coordinate frame with STalign
- Apply statistical testing with appropriate corrections
- Interpret results in biological context

Part 1: Introduction to Sample Integration

Why integrate? Understanding batch effects

Why Integrate Multiple Samples?

- Biological Questions:
  - Treatment vs control effects
  - Disease vs healthy tissue
  - Developmental time courses
  - Spatial replicates
  - Multi-region studies
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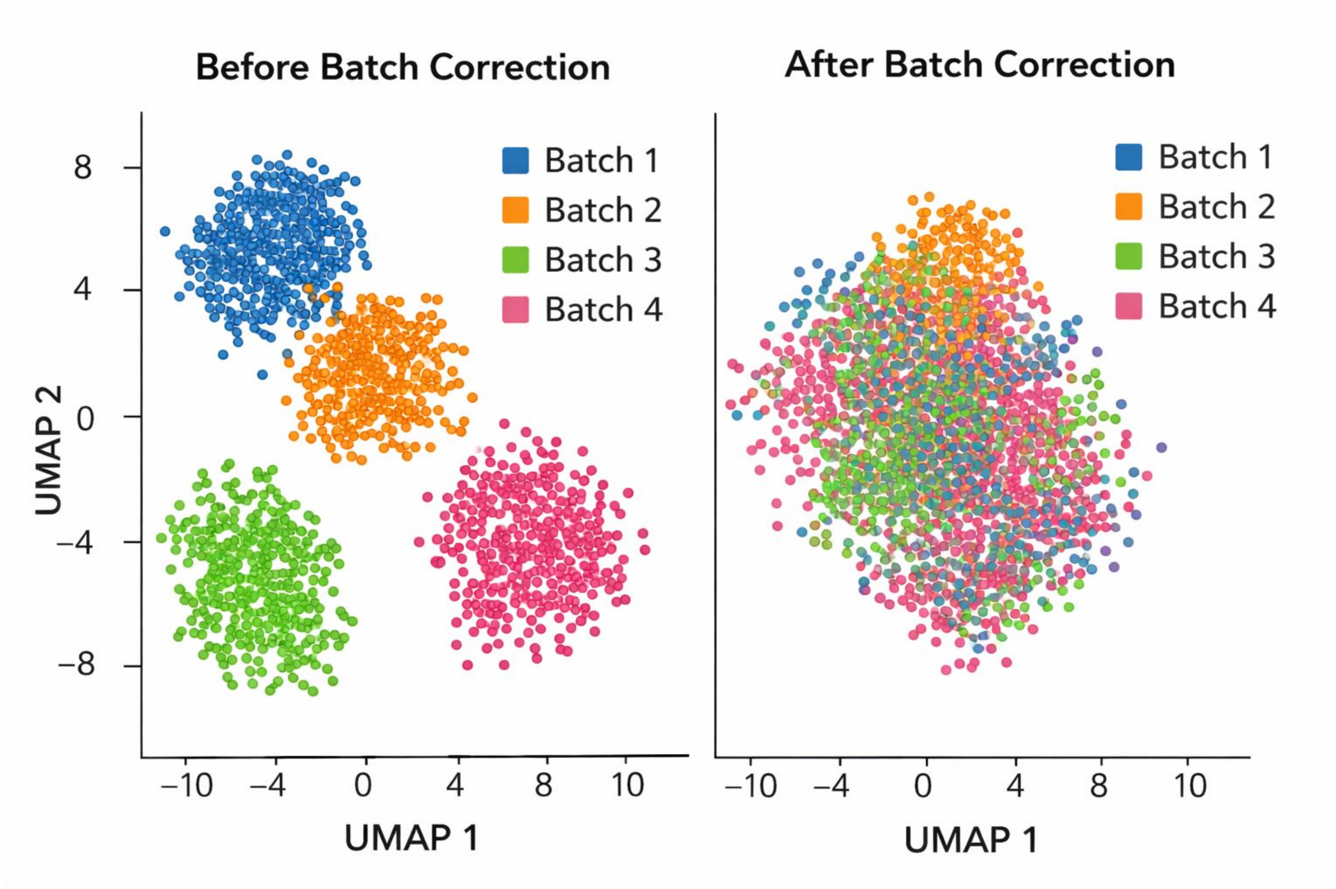
- Benefits of Integration:
  - Increased statistical power
  - More robust conclusions
  - Control for variability
  - Enable meta-analysis

- Technical Challenges:
  - Batch effects dominate signal
  - Sequencing depth differences
  - Processing date variations
  - Operator-specific effects
  - Platform differences
-
- Solution: Integration
  - Remove technical variation
  - Preserve biological signal
  - Enable fair comparison
  - Align samples in embedding space

Understanding Batch Effects

- What are batch effects?
    - Technical variations unrelated to biology
    - Can completely dominate biological signal
    - Cause spurious differential expression
-
- Common Sources:
    - Different processing dates or operators
    - Sequencing depth variations across runs
    - Library preparation batch differences
    - Reagent lot variations
    - Platform or protocol changes
- Detection:
    - Samples cluster by batch, not biology
    - Known markers show unexpected patterns
    - Strong PC correlates with batch ID



Part 2: Harmony Integration Algorithm

Fast, effective batch correction for spatial data

Harmony Algorithm Overview

- Key Features:
  - Works directly on PCA embeddings
  - Iterative soft clustering and correction
  - Handles multiple batches simultaneously
  - Computationally efficient and scalable
- Algorithm Steps:
  - 1. Cluster cells across all batches
  - 2. For each cluster, compute batch-specific centroids
  - 3. Move cells toward global centroid (soft assignment)
  - 4. Repeat until convergence
- Advantages:
  - Preserves cell type structure
  - Minimal parameter tuning needed
  - Much faster than MNN or Seurat
  - Works well with spatial data

Sample Integration Workflow

- Step 1: Preprocess individual samples
  - QC filtering, normalization, log-transform
  - Same pipeline for all samples
- Step 2: Concatenate samples
  - Combine into single AnnData object
  - Track sample/batch identity in metadata
- Step 3: Feature selection
  - Identify highly variable genes across all samples
  - Typically 3000-5000 genes
- Step 4: Compute PCA
  - Dimensionality reduction on combined data
  - Usually 50 principal components
- Step 5: Run Harmony
  - Correct PC coordinates using batch labels
  - Returns batch-corrected PCs
- Step 6: Downstream analysis
  - Use corrected PCs for neighbors, UMAP, clustering, etc.

Evaluating Integration Quality

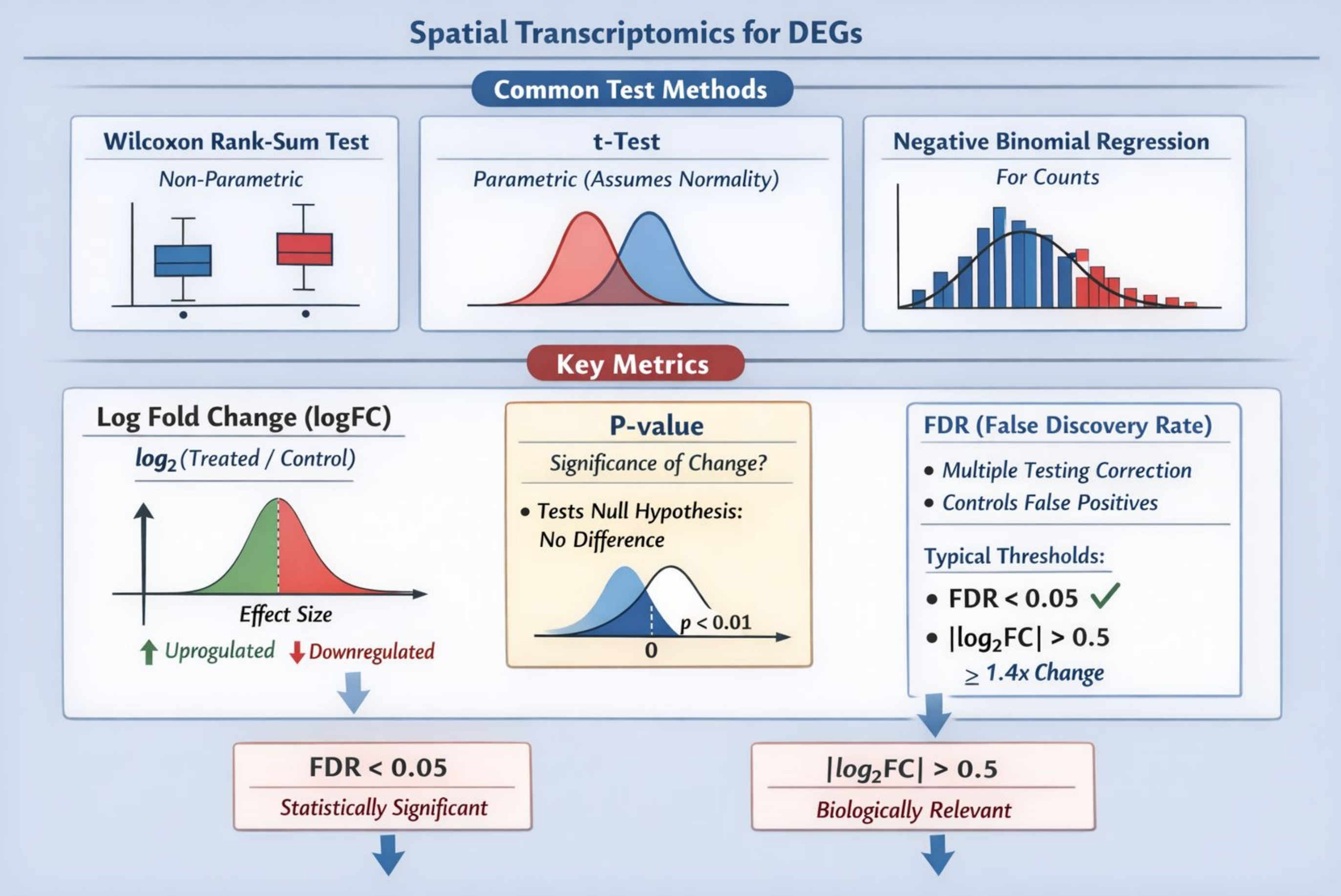
- Visual Inspection:
  - UMAP colored by batch (before/after)
  - UMAP colored by condition
  - UMAP colored by cell type markers
- Before Integration:
  - Samples separate by batch
  - Technical variation dominates
  - Cannot fairly compare conditions
- After Integration:
  - Batches mix within cell types
  - Biological structure preserved
  - Fair comparison possible
- Quantitative Metrics:
  - Batch mixing score (kBET)
  - Silhouette coefficient by cell type
  - Average Silhouette Width (ASW)
  - Local Inverse Simpson's Index (LISI)
- Warning Signs:
  - Over-correction
  - Loss of biological signal (metrics)
- Best Practice:
  - Always check known marker genes
  - Validate rare cell types survive
  - Compare multiple integration methods

Part 3: Differential Expression Analysis

Identifying genes that change between conditions

Statistical Testing for DEGs

- Common Test Methods:
  - Wilcoxon rank-sum test (non-parametric)
  - t-test (parametric, assumes normality)
  - Negative binomial regression (for counts)
- Log Fold Change (logFC):
  - Effect size - how much does expression change?
  - log2(Treated/Control)
  - Positive = upregulated, Negative = downregulated
  - |log2FC| > 0.5 (≈1.4x change, biological relevance)
- P-value:
  - Statistical significance - is change real?
  - Tests null hypothesis: no difference
- FDR (False Discovery Rate):
  - Multiple testing correction
  - Controls expected proportion of false positives
  - Typical Threshold: FDR < 0.05 (statistical significance)

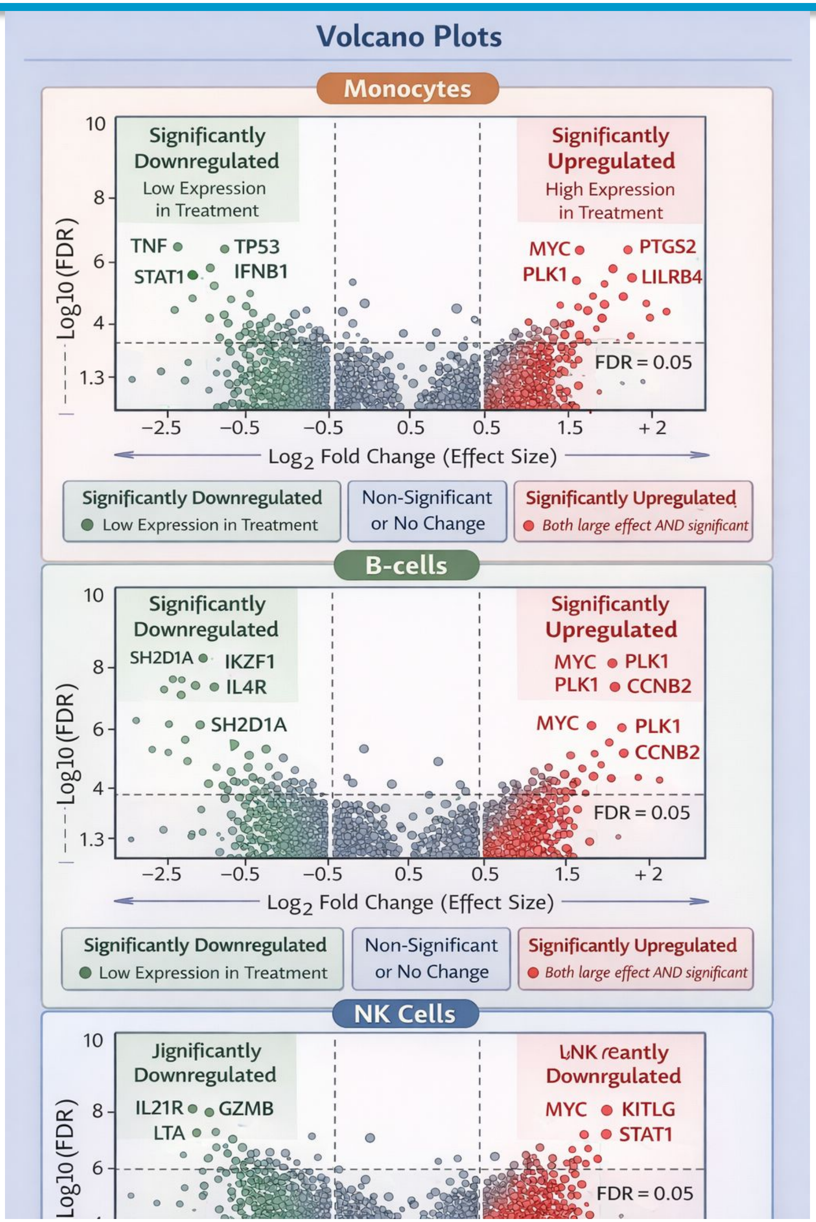


Which test to use?

Feature	t-test	Wilcoxon Rank-Sum
Parametric	Yes	No ✅
Requires normal distribution	Yes	No
Robust to outliers	No	Yes
Handles zero inflation	Poorly	Better
Works with small samples	Less reliable	Good
Metric used	Mean difference	Rank ordering

Volcano Plots: Visualizing Differential Expression

- Plot Structure:
  - X-axis: Log2 fold change (effect size)
  - Y-axis: -Log10(FDR) (significance)
  - Each point is one gene
- Top-right corner: Significantly upregulated
  - High expression in treatment - Both large effect AND significant
- Top-left corner: Significantly downregulated
  - Low expression in treatment - Both large effect AND significant
- Center-bottom: Non-significant or no change
- Best Practices:
  - Color-code by significance status
  - Label top genes for readability
  - Include threshold lines for clarity
  - Volcano plot per cell type

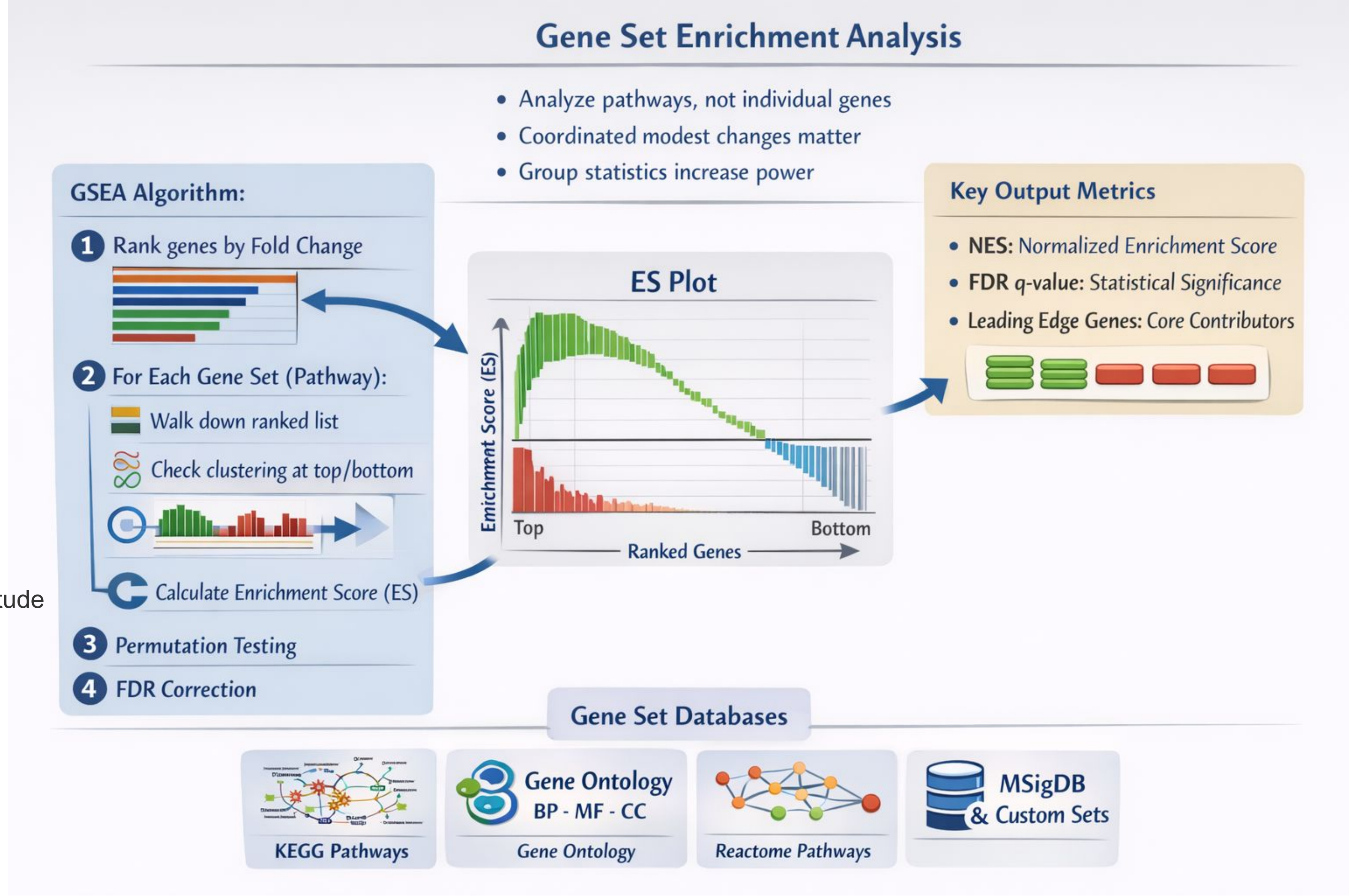


Part 4: Gene Set Enrichment Analysis

From genes to pathways - biological interpretation

Gene Set Enrichment Analysis (GSEA)

- Core Philosophy:
  - Analyze pathways, not individual genes
  - Coordinated modest changes matter
  - Group statistics increase power
- GSEA Algorithm:
  1. Rank all genes by fold change (or other metric)
  2. For each gene set (pathway):
     - - Walk down ranked list
     - - Check if gene set members cluster at top/bottom
     - - Calculate enrichment score (ES)
  3. Permutation testing for significance
  4. FDR correction across all gene sets
- Key Output Metrics:
  - NES (Normalized Enrichment Score): direction and magnitude
  - FDR q-value: statistical significance
  - Leading edge genes: core contributors
- Gene Set Databases:
  - KEGG pathways (metabolism, signaling)
  - Gene Ontology (BP, MF, CC)
  - Reactome, MSigDB, custom sets



KEGG Pathways for Biological Interpretation

- What is KEGG?
  - Kyoto Encyclopedia of Genes and Genomes
  - Curated biological pathway database
  - 500+ pathways covering all biology
  - Expert-curated from literature
- Pathway Categories:
  - Metabolism
    - Glucose, lipids, amino acids
  - Genetic Information
    - Transcription, translation
  - Environmental Information
    - Signal transduction
  - Cellular Processes
    - Cell cycle, apoptosis
- Using KEGG with GSEA:
  - Test enrichment for each pathway
  - Identify activated/suppressed processes
  - Generate testable hypotheses
- Example Interpretation: Finding: 20 upregulated, 30 downregulated genes
  - Hard to interpret individually
- GSEA reveals:
  - 'Oxidative phosphorylation' ↓ , 'Glycolysis' ↑
- Biological Story:
  - Warburg effect - metabolic switch
  - Cells shifting from mitochondrial to glycolytic metabolism
  - Coherent, testable hypothesis
- Power of Pathway Analysis:
  - Works even when individual genes not significant
  - 50 genes changing 1.3x each → significant pathway

Interpreting GSEA Results

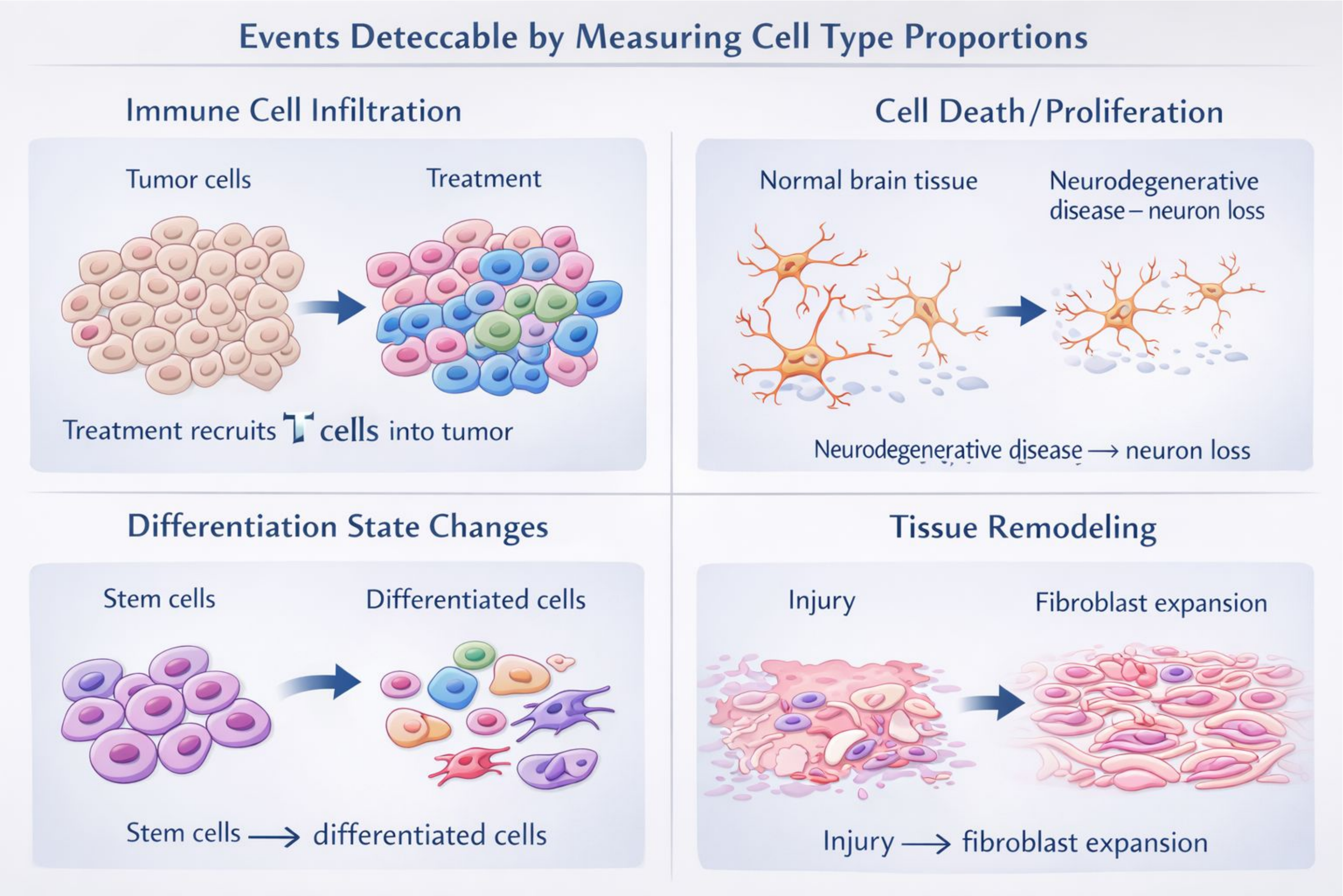
- NES (Normalized Enrichment Score):
  - Positive: pathway upregulated in condition
  - Negative: pathway downregulated
  - |NES| > 1.5 typically meaningful
- FDR q-value:
  - < 0.25: significant (more lenient than DEG)
  - < 0.05: highly significant
- Interpretation Strategy:
  - Look for coherent biological stories
  - Multiple related pathways changing together = strong signal
  - Context matters - same result can mean different things
  - Always check leading edge genes
- Example - Cancer Treatment:
  - Cell cycle pathways ↓ → Good (blocking proliferation)
  - Immune pathways ↑ → Good (immune activation)
  - DNA repair ↓ → Treatment working

Part 5: Cell Type Proportion Analysis

Does treatment change tissue composition?

Analyzing Cell Type Proportions

- Analysis Steps:
  - 1. Cluster integrated data into cell types
  - 2. Count cells per type per condition
  - 3. Calculate proportions (% of total)
  - 4. Statistical testing
  - 5. Visualize changes spatially
- Immune cell infiltration:
  - Treatment recruits T cells into tumor
- Cell death/proliferation:
  - Neurodegenerative disease → neuron loss
- Differentiation state changes:
  - Stem cells → differentiated cells
- Tissue remodeling:
  - Injury → fibroblast expansion
- Important Considerations:
  - Is change due to cell gain or other cell loss?
  - Check proliferation and apoptosis markers
  - Spatial patterns provide mechanism clues



Statistical Testing for Proportions

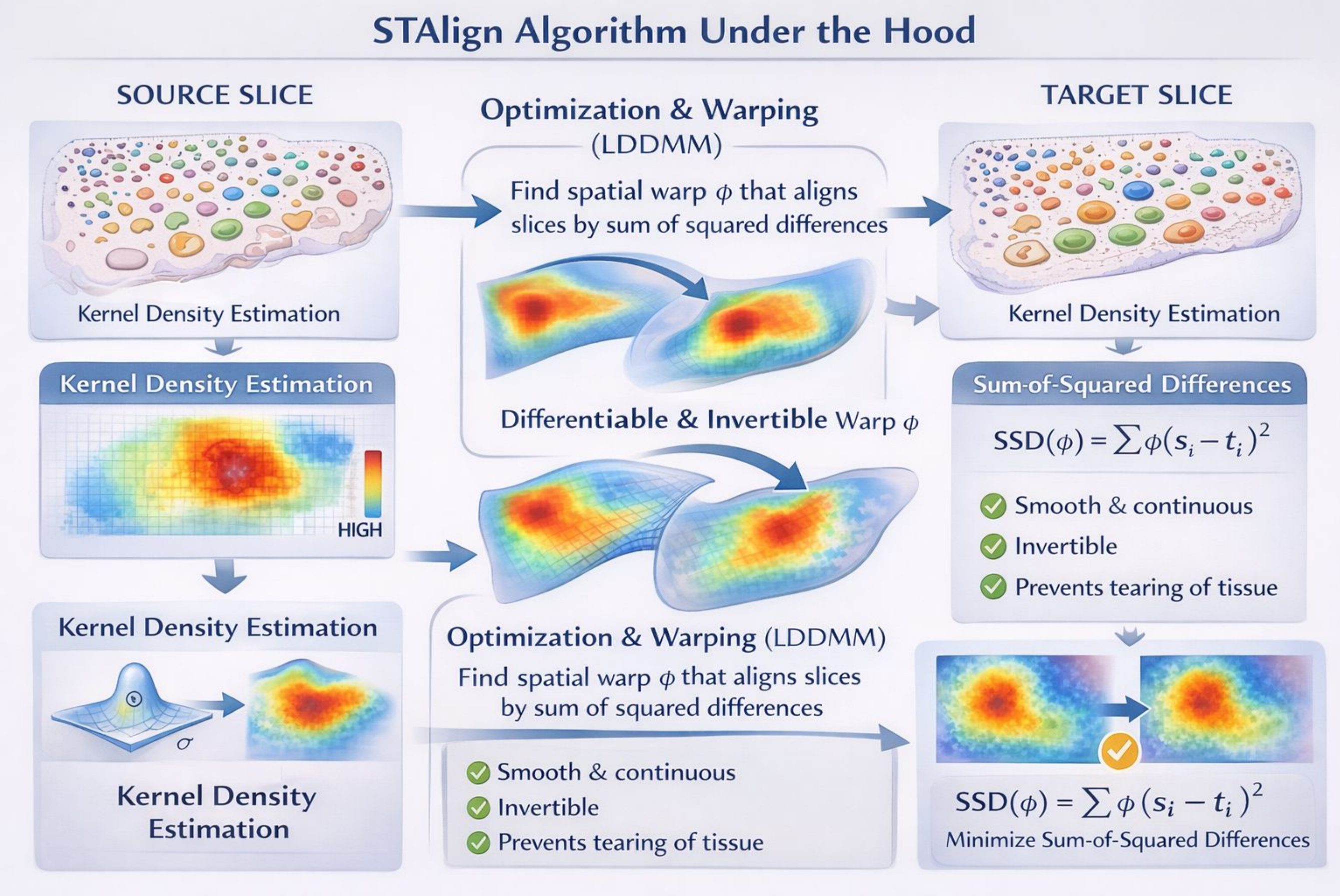
- Chi-square Test:
  - Tests overall distribution
  - Null: proportions same across conditions
  - Returns global p-value
- If significant:
  - At least one cell type differs
  - Doesn't tell you which one
- Requirements:
  - Sufficient cell counts
  - Independent observations
  - At least 5 cells per category
- Interpretation:
  - $p < 0.05$: distribution differs globally
  - Follow up with Fisher's tests
- Fisher's Exact Test:
  - Tests each cell type individually
  - 2×2 table: this type vs all others
  - Exact p-value (no approximation)
- Advantages:
  - Works with small counts
  - Identifies specific cell types
  - Provides odds ratios
- Multiple Testing Correction:
  - Test many cell types → many p-values
  - Use FDR (Benjamini-Hochberg)
  - Controls false discovery rate
- Odds Ratio Interpretation:
  - OR = 1: no enrichment
  - OR = 2: 2× more likely in treated
  - OR = 0.5: 2× less likely in treated

Part 6: Spatial Domain Alignment

Registering tissue slices with STalign

STalign Under the Hood: Problem Formulation

- The Alignment Problem:
  - Two slices: source S (x,y) and target T (x,y)
  - Coordinates are NOT comparable - different frames
  - Goal: find warp φ so that φ(S) ≈ T in physical space
- Tissue Representation:
  - Each slice is converted to a 2D spatial density field
  - Kernel density estimation over spot positions
  - Optional: weight by gene expression or cell-type probability
  - Two continuous images I_S(x,y) and I_T(x,y) to align
- Similarity Metric:
  - Sum-of-squared differences between density images
  - $||I_S(\varphi(x,y)) - I_T(x,y)||^2$ minimised over φ
  - Captures spatial overlap of tissue structure
- Framework: LDDMM
  - Large Deformation Diffeomorphic Metric Mapping
  - Allows large, smooth, invertible deformations
  - Standard tool in medical image registration

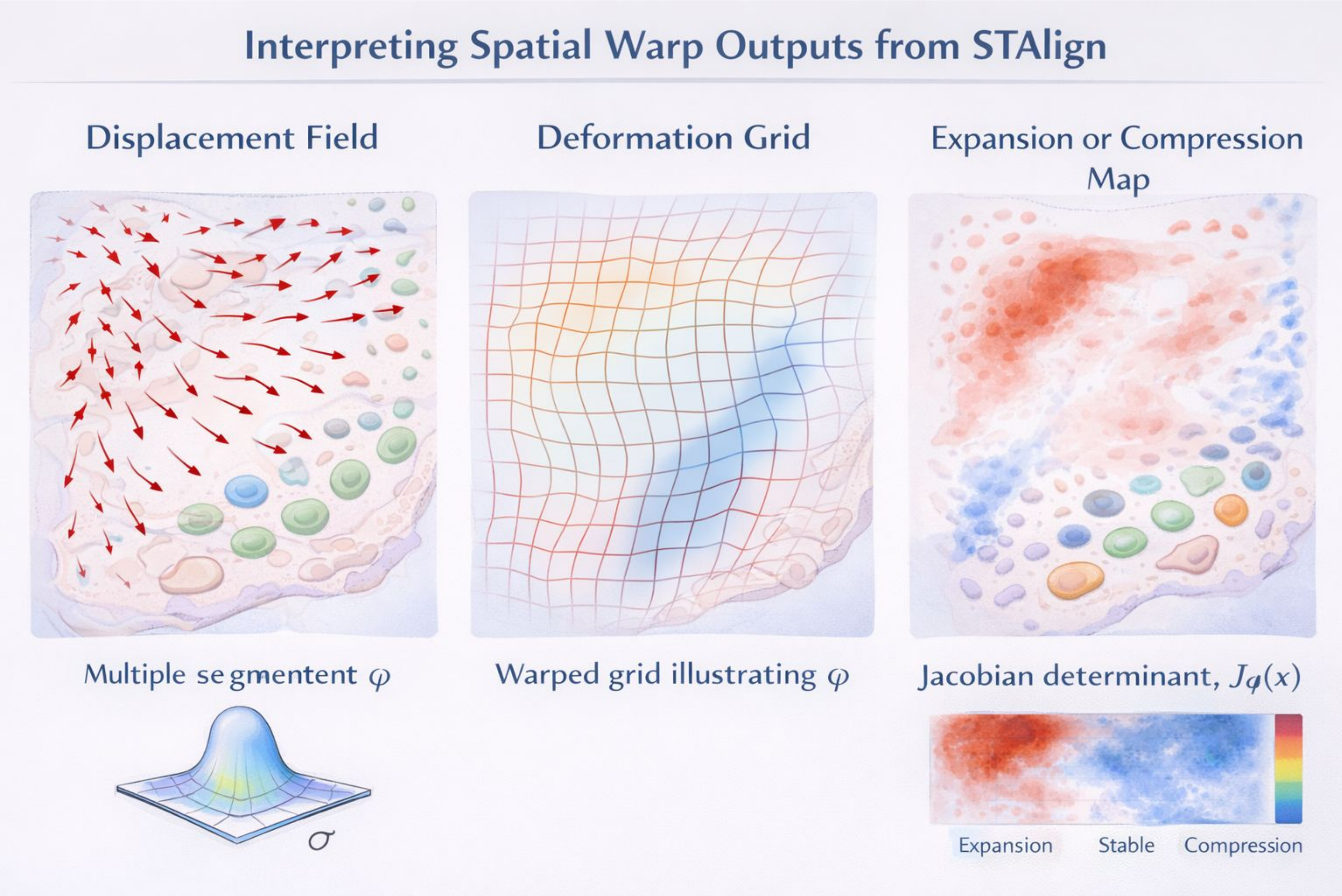


STalign Under the Hood: Diffeomorphic Optimisation

- Velocity Field Parameterisation:
  - Warp φ is NOT specified directly
  - Instead: define time-varying velocity field $v(x, t)$
  - Integrate ODE $dx/dt = v(x,t)$ from $t=0$ to $t=1$
  - φ = flow of $v \rightarrow$ always diffeomorphic
- Energy Functional:
  - $E(v)$ = Data term + $\lambda \cdot$ Regularisation term
  - Data: $||I\_S(\varphi) - I\_T||^2$ (alignment quality)
  - Reg: $||Lv||^2$ (smoothness of deformation)
  - L = differential operator (Laplacian-based)
  - λ controls rigidity vs. flexibility trade-off
- Optimisation:
  - Gradient descent on velocity field parameters
  - Gradient computed via adjoint method (backprop)
  - Scales to millions of spots via GPU acceleration
- Output & What to Do With It:
  - Transformed coordinates (x', y') per spot
  - Store in adata.obsm['spatial_aligned']
  - Both slices now live in same coordinate frame
- Downstream Analyses Enabled:
  - Cell-type domain correspondence across sections
  - Region-level DEG analysis (matched spatial zones)
  - Morphological overlay plots
  - Inter-slice cell-communication inference

Spatial Visualization of Treatment Effects

- Why Spatial Matters:
  - See WHERE changes occur in tissue
  - Identify regional vs global effects
  - Understand tissue architecture
  - Generate mechanism hypotheses
- Side-by-side comparison:
  - Control vs Treated, same gene/cell type
  - Use identical color scales
- Gene expression maps:
  - Show DEGs in spatial context
- Cell type distributions:
  - Overlay cluster assignments on tissue
- Statistical overlays:
  - Color by fold change or p-value
- Spatial Patterns Reveal Mechanism:
  - Boundary effects → local processes at interfaces
  - Regional specificity → zone-specific responses
  - Gradients → diffusible signals
  - Hotspots → local production or aggregation



Best Practices & Common Pitfalls

- Always Do:
  - ✓ Visually inspect integration quality
  - ✓ Use biological replicates if possible
  - ✓ Set appropriate statistical thresholds
  - ✓ Apply multiple testing correction
  - ✓ Validate top DEGs independently
  - ✓ Check for known marker genes
  - ✓ Visualize results spatially
  - ✓ Interpret in biological context
  - ✓ Report all methods and parameters
  - ✓ Compare multiple integration methods
- Never Do:
  - ✗ Skip visual quality control
  - ✗ Compare without integration
  - ✗ Ignore batch effects
  - ✗ Report uncorrected p-values
  - ✗ Cherry-pick favorable pathways
  - ✗ Over-interpret borderline results
  - ✗ Trust outlier genes without validation
  - ✗ Forget biological plausibility
  - ✗ Lower thresholds to get 'significance'
  - ✗ Skip replication if possible

Hands-on Exercise: Complete Workflow

- Dataset:
  - Control: adult_mouse_brain_ST4k.h5ad
  - Treated: adult_mouse_brain_ST4k_treated.h5ad (Mock treatment effects simulated)
- Complete Analysis Tasks:
  - 1. Load and QC both samples
  - 2. Integrate with Harmony
  - 3. Identify differentially expressed genes
  - 4. Create volcano plot
  - 5. Run GSEA with KEGG pathways
  - 6. Analyze cell type proportion changes
  - 7. Visualize top DEGs spatially
  - 8. Align slices with STalign and compare spatial domains
  - 9. Create publication-quality figures
- Tools & Resources:
  - Jupyter notebook: sample_integration.ipynb
  - Python packages: scanpy, harmonypy, gseapy, stalign
  - Step-by-step guidance provided

Key Takeaways

- Integration removes batch effects while preserving biology
  - Essential before comparative analysis
- Harmony is fast and effective for spatial data
  - Works in PCA space, handles multiple batches
- Differential expression identifies WHAT changed
  - Use appropriate thresholds and FDR correction
- GSEA reveals WHY - biological pathways affected
  - More interpretable than gene lists alone
- Proportion analysis detects compositional changes
  - Important complement to DEG analysis
- STalign registers slices to a shared coordinate frame
  - Diffeomorphic warp preserves tissue topology
  - Enables direct region-to-region cross-sample comparison
- Spatial visualization shows WHERE changes occur
  - Provides mechanistic insights
- Always validate integration quality
  - Visual inspection is critical

Additional Resources

- Software & Documentation:
  - Scanpy: https://scanpy.readthedocs.io
  - Harmonypy: https://github.com/slowkow/harmonypy
  - GSEApy: https://gseapy.readthedocs.io
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- Pathway Databases:
  - KEGG: https://www.genome.jp/kegg/
  - Gene Ontology: http://geneontology.org
  - MSigDB: https://www.gsea-msigdb.org
  - Reactome: https://reactome.org
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- Key Publications:
  - Harmony: Korsunsky et al. Nature Methods 2019
  - GSEA: Subramanian et al. PNAS 2005
  - Spatial transcriptomics review: Moses & Pachter Nat Rev Genet 2022

Questions?

Thank you!